

# DIGITAL FORENSIC EXAMINATION REPORT

National Technical Research Organisation (NTRO) — Case Reference: **CASE-2026-001**

|                     |                                                                 |                         |                      |
|---------------------|-----------------------------------------------------------------|-------------------------|----------------------|
| Case Number:        | CASE-2026-001                                                   | Target Host:            | WS-FIN-04.fin.local  |
| Case Title:         | Suspicious Execution & Exfiltration on Workstation<br>WS-FIN-04 | Status:                 | ACTIVE               |
| Lead Investigator:  | Lead DFIR Analyst<br>(NTRO-INV-42)                              | Date & Time:            | 2026-09-26 11:18 UTC |
| Forensic Framework: | NTRO DSL-DFIR v2.4 (Authorized Mode)                            | Evidence Hash Protocol: | FIPS 180-4 SHA-256   |

## 1. EXECUTIVE SUMMARY

An authorized forensic investigation was commenced for target host **WS-FIN-04.fin.local** under NTRO digital evidence procedures. Forensic analysis utilized non-intrusive domain-specific scripting language (Forensic DSL) commands operating under **FORENSIC\_READ\_ONLY** policies. A total of **25** evidence items were acquired with verified SHA-256 cryptographic digests. The deterministic rule engine identified **8** notable security findings.

## 2. KEY SUSPICIOUS FINDINGS

| Severity | Rule & Title                                                                                                                                                             | Category | Evidence Reference                                     |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------|
| CRITICAL | <b>IOC Hash Match: CobaltStrike Beacon Synthetic Stager</b><br>File 'update_svc_helper.exe' (C:\Users\jdoe\AppData\Local\Temp\update_svc_helper.exe) matc...             | IOC      | C:\Users\jdoe\AppData\Local\Temp\update_svc_helper.exe |
| HIGH     | <b>Executable in Monitored User Directory: update_svc_helper.exe</b><br>Executable binary or script discovered residing in high-risk directory 'C:\Users\jdoe\AppData... | FILE     | C:\Users\jdoe\AppData\Local\Temp\update_svc_helper.exe |
| HIGH     | <b>Executable in Monitored User Directory: payload_stager.ps1</b><br>Executable binary or script discovered residing in high-risk directory 'C:\Users\jdoe\Down...       | FILE     | C:\Users\jdoe\Downloads\payload_stager.ps1             |
| HIGH     | <b>Suspicious Process Execution: update_svc_helper.exe (PID:4820)</b><br>Process 'update_svc_helper.exe' running from temporary path 'C:\Users\jdoe\AppData\Local\T...   | PROCESS  | PID-4820                                               |

|                 |                                                                                                                                                              |         |                                           |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------------------------------------------|
| <b>HIGH</b>     | <b>Obfuscated PowerShell Invocation Detected (PID:2104)</b><br>PowerShell process invoked with evasion/download flags: powershell.exe -NonI -W Hidden -Ex... | PROCESS | PID-2104                                  |
| <b>CRITICAL</b> | <b>IOC Outbound Network C2 Match: 198.51.100.45:4444</b><br>Active socket initiated by update_svc_helper.exe (PID:4820) connects to known Threat Intel...    | NETWORK | 192.168.1.105:51442 -> 198.51.100.45:4444 |
| <b>CRITICAL</b> | <b>IOC Outbound Network C2 Match: 203.0.113.88:8080</b><br>Active socket initiated by powershell.exe (PID:2104) connects to known Threat Intel IP 203...     | NETWORK | 192.168.1.105:52110 -> 203.0.113.88:8080  |
| <b>MEDIUM</b>   | <b>Authentication Failure Anomaly (1 event(s) recorded)</b><br>Detected authentication failure events. Detailed probe against workstation credentials....    | LOG     | Security Log Event 4625                   |

### 3. EVIDENCE INVENTORY & SHA-256 HASH VERIFICATION

| ID              | Artifact Name                          | Type   | SHA-256 Digest                  | Integrity |
|-----------------|----------------------------------------|--------|---------------------------------|-----------|
| EVD-CASE-20-001 | System_Config_WS-FIN-04.fin.local.json | SYSTEM | e3b0c44298fc1c149afb4c8996f...  | VERIFIED  |
| EVD-CASE-20-002 | update_svc_helper.exe                  | FILE   | 8f4e2a6d5c1b9e0f3d7a8b4c2e6f... | VERIFIED  |
| EVD-CASE-20-003 | payload_stager.ps1                     | FILE   | c8a49c2d1b5e3f7a9d0b2c4e6f8a... | VERIFIED  |
| EVD-CASE-20-004 | quarterly_budget_2026.xlsx             | FILE   | 1a2b3c4d5e6f7a8b9c0d1e2f3a4b... | VERIFIED  |
| EVD-CASE-20-005 | invoice_q3_confidential.pdf            | FILE   | 7d8e9f0a1b2c3d4e5f6a7b8c9d0e... | VERIFIED  |
| EVD-CASE-20-006 | System_Config_WS-FIN-04.fin.local.json | SYSTEM | e3b0c44298fc1c149afb4c8996f...  | VERIFIED  |
| EVD-CASE-20-007 | update_svc_helper.exe                  | FILE   | 8f4e2a6d5c1b9e0f3d7a8b4c2e6f... | VERIFIED  |
| EVD-CASE-20-008 | payload_stager.ps1                     | FILE   | c8a49c2d1b5e3f7a9d0b2c4e6f8a... | VERIFIED  |
| EVD-CASE-20-009 | quarterly_budget_2026.xlsx             | FILE   | 1a2b3c4d5e6f7a8b9c0d1e2f3a4b... | VERIFIED  |
| EVD-CASE-20-010 | invoice_q3_confidential.pdf            | FILE   | 7d8e9f0a1b2c3d4e5f6a7b8c9d0e... | VERIFIED  |
| EVD-CASE-20-011 | System_Config_WS-FIN-04.fin.local.json | SYSTEM | e3b0c44298fc1c149afb4c8996f...  | VERIFIED  |

|                      |                                        |        |                                 |          |
|----------------------|----------------------------------------|--------|---------------------------------|----------|
| EVD-CASE-20<br>2-012 | update_svc_helper.exe                  | FILE   | 8f4e2a6d5c1b9e0f3d7a8b4c2e6f... | VERIFIED |
| EVD-CASE-20<br>2-013 | payload_stager.ps1                     | FILE   | c8a49c2d1b5e3f7a9d0b2c4e6f8a... | VERIFIED |
| EVD-CASE-20<br>2-014 | quarterly_budget_2026.xlsx             | FILE   | 1a2b3c4d5e6f7a8b9c0d1e2f3a4b... | VERIFIED |
| EVD-CASE-20<br>2-015 | invoice_q3_confidential.pdf            | FILE   | 7d8e9f0a1b2c3d4e5f6a7b8c9d0e... | VERIFIED |
| EVD-CASE-20<br>2-016 | System_Config_WS-FIN-04.fin.local.json | SYSTEM | e3b0c44298fc1c149afb4c8996f...  | VERIFIED |
| EVD-CASE-20<br>2-017 | update_svc_helper.exe                  | FILE   | 8f4e2a6d5c1b9e0f3d7a8b4c2e6f... | VERIFIED |
| EVD-CASE-20<br>2-018 | payload_stager.ps1                     | FILE   | c8a49c2d1b5e3f7a9d0b2c4e6f8a... | VERIFIED |
| EVD-CASE-20<br>2-019 | quarterly_budget_2026.xlsx             | FILE   | 1a2b3c4d5e6f7a8b9c0d1e2f3a4b... | VERIFIED |
| EVD-CASE-20<br>2-020 | invoice_q3_confidential.pdf            | FILE   | 7d8e9f0a1b2c3d4e5f6a7b8c9d0e... | VERIFIED |
| EVD-CASE-20<br>2-021 | System_Config_WS-FIN-04.fin.local.json | SYSTEM | e3b0c44298fc1c149afb4c8996f...  | VERIFIED |
| EVD-CASE-20<br>2-022 | update_svc_helper.exe                  | FILE   | 8f4e2a6d5c1b9e0f3d7a8b4c2e6f... | VERIFIED |
| EVD-CASE-20<br>2-023 | payload_stager.ps1                     | FILE   | c8a49c2d1b5e3f7a9d0b2c4e6f8a... | VERIFIED |
| EVD-CASE-20<br>2-024 | quarterly_budget_2026.xlsx             | FILE   | 1a2b3c4d5e6f7a8b9c0d1e2f3a4b... | VERIFIED |
| EVD-CASE-20<br>2-025 | invoice_q3_confidential.pdf            | FILE   | 7d8e9f0a1b2c3d4e5f6a7b8c9d0e... | VERIFIED |

#### 4. CHAIN OF CUSTODY AUDIT TRAIL

| Timestamp           | Action            | Investigator      | Details & Chain Digest                                                                                       |
|---------------------|-------------------|-------------------|--------------------------------------------------------------------------------------------------------------|
| 2026-09-26T11:18:03 | EVIDENCE_ACQUIRED | Lead DFIR Analyst | Acquired update_svc_helper.exe (FILE). Initial SHA-256 calculated: 8f4...<br>Digest: 25e9841ff588f66a0364... |
| 2026-09-26T11:18:03 | EVIDENCE_ACQUIRED | Lead DFIR Analyst | Acquired payload_stager.ps1 (FILE). Initial SHA-256 calculated: c8a49c...<br>Digest: a76bba56945c95dbd01e... |
| 2026-09-26T11:18:03 | EVIDENCE_ACQUIRED | Lead DFIR Analyst | Acquired quarterly_budget_2026.xlsx (FILE). Initial SHA-256 calculated...<br>Digest: a5c5158903f734729666... |

|                     |                             |                   |                                                                                                                 |
|---------------------|-----------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------|
| 2026-09-26T11:18:03 | EVIDENCE_ACQUIRE<br>D       | Lead DFIR Analyst | Acquired invoice_q3_confidential.pdf (FILE).<br>Initial SHA-256 calculate...<br>Digest: d76eeb941db3e8657239... |
| 2026-09-26T11:18:03 | TIMELINE_CORRELA<br>TED     | Lead DFIR Analyst | Reconstructed and correlated unified<br>chronological timeline comprising...<br>Digest: a65f17e874edabd284b6... |
| 2026-09-26T11:18:03 | RULE_ANALYSIS_CO<br>MPLETED | Lead DFIR Analyst | Deterministic rule engine evaluation<br>completed: 8 suspicious findings ...<br>Digest: f9d380099e4623a4dded... |

## 5. INVESTIGATOR ATTESTATION & CONCLUSION

I hereby attest that the digital evidence items cataloged in this report were acquired and processed in accordance with standardized forensic procedures. All operations executed strictly within read-only boundaries under authorized cryptographic control. No system state was illegally modified.

**Examiner:** Lead DFIR Analyst

**Badge:** NTRO-INV-42

**Signature:** [CRYPTOGRAPHICALLY SEALED BY  
NTRO-INV-42]

**Date:** 2026-09-26 11:18:03 UTC